3D Printing Filaments

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
PLA Filament; ABS Filament; PETG Filament; Nylon Filament; TPU Filament	IS 14434	ISO/ASTM 52900:2021 — AM fundamentals & terminology; ISO 17296-2:2015 — AM process categories/feedstock; ISO/ASTM 52927:2024 — AM test/quality framework; ISO 527-1 / ISO 527-2 — Tensile properties; ISO 1133-1 — Melt mass-flow rate; ISO 178 — Flexural properties; ISO 179-1 — Charpy impact; ISO 180 — Izod impact; ISO 75 — Deflection temperature under load; ISO 306 — Vicat softening temperature; ISO 3451-1 — Determination of ash; ISO 11357-1 / ISO 11357-3 — DSC / melting & crystallization; ISO 4892-3 — Artificial weathering, where applicable.	ASTM D638 — Tensile properties; ASTM D790 — Flexural properties; ASTM D256 — Izod impact; ASTM D3418 — DSC/thermal transitions; ASTM D1238 — Melt flow rates of thermoplastics; ASTM D648 — Deflection temperature under load; ASTM D1525 — Vicat softening temperature; ASTM D570 — Water absorption; ASTM D792 — Density/specific gravity; ASTM D2863 — Limiting oxygen index, where flammability characterization is required; ASTM E1131 — Compositional/thermal analysis by thermogravimetry, where relevant.	DIN EN ISO 527-1 / 527-2 — Tensile; DIN EN ISO 1133-1 — Melt mass-flow rate; DIN EN ISO 178 — Flexural; DIN EN ISO 179-1 — Charpy impact; DIN EN ISO 180 — Izod impact; DIN EN ISO 75 — HDT; DIN EN ISO 306 — Vicat; DIN EN ISO 4892-3 — Artificial weathering, where applicable; DIN EN ISO 11357 series — DSC/thermal analysis.	UL 94 — Flammability classification, where required; FDA 21 CFR 177 — Food-contact applications, where applicable; USP Class VI — Medical applications, where applicable
PEEK Filament	-	ISO/ASTM 52900:2021 — AM fundamentals; ISO 23153-1:2020 — PEEK moulding/extrusion material designation; ISO 527-1 / 527-2 — Tensile; ISO 1133-1 — Melt flow; ISO 178 — Flexural; ISO 179-1 — Charpy impact; ISO 75 — HDT; ISO 306 — Vicat; ISO 11357 series — DSC/thermal analysis; ISO 3451-1 — Ash/filler content, where relevant.	ASTM D8033 — PEEK moulding/extrusion materials; ASTM D638 — Tensile; ASTM D790 — Flexural; ASTM D3418 — Thermal properties; ASTM D1238 — Melt flow; ASTM D648 — HDT; ASTM D1525 — Vicat; ASTM D570 — Water absorption; ASTM D792 — Density. ASTM D6262 — only for PAEK basic shapes, so not a primary filament standard.	DIN EN ISO 527-1 / 527-2 — Tensile; DIN EN ISO 1133-1 — Melt flow; DIN EN ISO 178 — Flexural; DIN EN ISO 179-1 — Impact; DIN EN ISO 75 — HDT; DIN EN ISO 306 — Vicat; DIN EN ISO 11357 series — DSC.	UL 94 — Flammability, conditional; FDA 21 CFR 177 — Food-contact application, conditional; USP Class VI — Medical application, conditional
Carbon Fibre Reinforced Filament	-	ISO/ASTM 52900:2021 — AM fundamentals; ISO 527-4 — Tensile properties of isotropic/orthotropic fibre-reinforced plastics; ISO 527-5 — Unidirectional fibre-reinforced plastics, only where applicable; ISO 14125 — Flexural properties of fibre-reinforced plastics; ISO 1172 — Glass/carbon textile composite ignition method, where applicable; ISO 3451-1 — Ash/filler content, where applicable; ISO 11357 series — Polymer thermal analysis.	ASTM D3039/D3039M — Composite tensile testing, conditional on specimen construction; ASTM D7264/D7264M — Composite flexural properties; ASTM D3171 — Constituent content; ASTM D3418 — Polymer thermal properties; ASTM D792 — Density; ASTM D2584 — Ignition loss for composite reinforcement content, where applicable.	DIN EN ISO 527-4 — Fibre-reinforced plastics tensile; DIN EN ISO 527-5 — Unidirectional fibre-reinforced plastics; DIN EN ISO 14125 — Composite flexural; DIN EN ISO 178 — Flexural testing; DIN EN ISO 179-1 — Impact; DIN EN ISO 11357 series — DSC/thermal characterization.	UL 94 — Flammability, where required; SAE AMS specifications — only for the specific aerospace material/application, not generic carbon-fibre filament

Source field: Plastic and Plastic Products